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Guangdong Province's 14th Five-Year Plan for Addressing Climate Change

July 2022

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Preface

Climate change is a common challenge for all mankind. Responding to climate change is related to the sustainable development of the Chinese nation and the overall economic and social development of our country. General Secretary Xi Jinping has repeatedly stressed that We should not be asked to deal with climate change, but we should do it ourselves. It is an inherent requirement for sustainable development and also a responsibility to promote the building of a community with a shared future for mankind. To address climate change, we must actively control greenhouse gas emissions and slow down climate change trends. We should focus on reducing the adverse impacts of climate change on the natural ecological, economic and social fields. Mitigation and adaptation are two major strategies for addressing climate change. They complement each other and are indispensable. Can.

Guangdong is China's largest economic province, but also a major energy and resource consumer and a major climate, Provinces with greater environmental pressure should actively respond to climate change and achieve low-carbon development. Protect the province's economic security, energy security, ecological security, food security and people's lives Property safety is of vital importance. The 14th Five-Year Plan period is the beginning of my country's comprehensive construction of a socialist The first five years of Guangdong's new journey towards becoming a modern socialist country are a period of great efforts to build a socialist The first one to take the lead in the country and create new glories in the new journey of building a modern socialist country The past five years have also been a period for Guangdong to steadily move towards the goals of “carbon peak and carbon neutrality” and lay a solid foundation. In order to implement the CPC Central Committee and the State Council's work on addressing climate change Decision-making and deployment, thoroughly implement General Secretary Xi Jinping's series of important speeches and important instructions to Guangdong In accordance with the spirit of the instructions, the Fourteenth Five-Year Plan for National Economic and Social Development of Guangdong Province The Outline of the 2035 Vision Goals and the 2035 Plan for Addressing Climate Change in Guangdong Province was compiled.

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The 14th Five-Year Plan (hereinafter referred to as the "Plan") puts forward the guiding ideology, basic principles and development goals for our province to respond to climate change during the 14th Five-Year Plan period, and clarifies the

The Plan will serve as the “Ten

It is an important basis and action guide for promoting the province's work on climate change during the "four-five" period and even longer, and it will help Guangdong explore a coordinated path of carbon reduction, pollution reduction, green expansion and growth. Adjusting the development path provides solid guarantee.

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Chapter 1 Current Situation and Situation

Section 1 Climate Change Trends and Impacts

Against the backdrop of global warming, Guangdong Province's climate has also undergone significant changes.

Since 1961, the average temperature in Guangdong Province has risen by 0.19°C every 10 years, which is higher than the global average.

The annual precipitation and total water resources in the province are

There is a small cycle of alternating wet and dry weather around the normal value. Typhoons, rainstorms, floods, extreme highs

The frequency of meteorological disasters such as high temperatures and periodic droughts has increased.

Climate change has adverse impacts on the province's natural ecosystems, economy and society.

From 1980 to 2020, the average annual rate of sea level rise in the coastal areas of Guangdong Province is 3.5 mm/

In 2016, the sea level rise was higher than the national average level in the coastal areas during the same period.

Disaster severity of storm surges, coastal erosion and saltwater intrusion along the coast of Guangdong Province, 2016-2020

The average direct economic loss from marine disasters in the province is 1.76 billion yuan per year. The number of high temperature days increases.

The energy consumption for cooling in summer generally increases, and the uneven temporal and spatial distribution of precipitation has a great impact on hydropower generation.

The impact of the increase in energy supply in the province has increased.

In addition, the spring phenological period of plants tends to be earlier, and agricultural production faces drought and insect pest risks.

The risk of agricultural output instability is increasing.

Floods occur from time to time, super typhoons cause coastal tides to hit new highs, and regional drought risks increase

In addition, the pressure on the safe and stable construction and operation of urban lifelines, urban and rural infrastructure, and engineering projects has increased. High

temperature heat waves, heat island effects and other phenomena have intensified, and high temperature heat stroke, infectious diseases, and other diseases have increased.

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The risk of disease increases and the threat to human health increases.

Section 2 Work Results

Since the 13th Five-Year Plan, our province has given full play to green and low-carbon development to lead the construction of ecological civilization actively explore low-carbon development systems and mechanisms, and make solid progress in addressing climate change.

During the 13th Five-Year Plan period, carbon intensity decreased by 22.35%, exceeding the national target. It has been rated “excellent” in the national carbon intensity assessment for many consecutive years.

The level of green and low-carbon development continues to improve.

The industrial structure with leading, advanced manufacturing and modern service industries as the main body is expected to be The added value of the three industries accounts for 4.3:39.2:56.5, and the secondary industry accounts for the regional GDP. The proportion of the tertiary industry has gradually decreased, the proportion of the secondary industry has increased year by year, and the added value of the secondary industry has increased year by year. Carbon emissions dropped from 1.037 tons/10,000 yuan in 2015 to 0.772 tons/10,000 yuan in 2020. The average annual decrease was 5.7%, with a cumulative decrease of 25.5%. The energy structure has been significantly optimized. In 2020, all The total energy consumption of the province is 345 million tons of standard coal, including coal products, oil products, natural gas, and other energy products. The proportion of sources (including primary electricity) is 33.4%, 26.2%, 9.8% and 30.6%, and non-fossil Energy accounts for 30.3% of total energy consumption. Compared with 2015, coal consumption has The proportion of natural gas consumption increased by about 3.5 percentage points, and the proportion of non-chemical The proportion of fossil energy consumption increased by 5.7 percentage points.

The low-carbon innovation demonstration pilot project was effectively completed. The carbon emission trading pilot project was running smoothly. As of December 31, 2021, the province's cumulative carbon emission quota trading volume reached 199.7 million tons, with a cumulative trading amount of 4.61 billion yuan, ranking first in the national regional carbon market.

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The pilot program of the Generalized System of Preferences was launched, and the first carbon preference platform in China was launched, which incorporated carbon preference emission reduction projects into the carbon emission rights trading system. The carbon capture, utilization and storage demonstration project achieved phased results. The first power plant in Asia and the third in the world were put into operation at China Resources (Haifeng) Power Plant in Shenzhen-Shantou Cooperation Zone. Actively promote the construction of national low-carbon city pilots in Guangzhou, Shenzhen and Zhongshan, and promote the pilot projects of near-zero carbon emission zone demonstration projects and low-carbon

Many demonstration projects have been carried out, including industrial parks, low-carbon communities, low-carbon product certification and carbon labeling.

The reform of the system and mechanism is gradually improving. The policy system for addressing climate change is gradually improving.

The Implementation Plan for Controlling Greenhouse Gas Emissions during the 13th Five-Year Plan Period of Guangdong Province and The 13th Five-Year Plan for Addressing Climate Change in Dongguan Province and other documents have established

The above-mentioned cities have established a normalized working mechanism for compiling and reporting greenhouse gas emission inventories. Guangdong Province's major engineering construction project headquarters coordination mechanism, disaster prevention and mitigation "four ones" The construction of emergency response mechanism, leading group for responding to climate change and energy conservation and emission reduction, etc.

The ability to adapt to climate change is constantly improving. The province is vigorously promoting the construction of sponge cities. According to self-assessment by various regions, the proportion of sponge city construction areas that meet the standards in the province is about 23.72%. The ecological protection red line was adjusted, and the forest coverage rate in the province reached 58.66%. With the improvement of management capacity, the proportion of meteorological disaster losses in GDP has dropped from 0.36% to 0.19%.

Section 3 Development Situation

During the 14th Five-Year Plan period, the province's ecological civilization construction has entered a stage of coordinated growth with pollution reduction and carbon reduction. At this critical period when we must take efficiency as the key point, promote the comprehensive green transformation of economic and social development, and connect the vision of carbon peak and carbon neutrality, the work of responding to climate change faces major opportunities.

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General Secretary Xi Jinping's important public oath has set a great precedent for our province in addressing climate change and achieving carbon reduction. Peak carbon neutrality provides a strong ideological and action guide. During the 14th Five-Year Plan period, Our province is at the forefront of the country in the new journey of building a modern socialist country in an all-round way. The first five years of creating new glories is an important window to show the world the effectiveness of our province's work in addressing climate change. As the forefront of reform and opening up, Guangdong will make greater efforts to Support the timely completion of the carbon peak and carbon neutrality goals and promote the formation of work to address climate change New pattern.

Technological and industrial innovations are accelerating the economic development into a green transformation opportunity period. A round of scientific and technological revolution and industrial transformation has been in-depth, promoting the optimization and upgrading of Guangdong Province's industrial structure. level, improve resource and energy utilization efficiency, accelerate the green transformation of manufacturing industry, green and low-carbon The spatial pattern, industrial structure, production and lifestyle of the economy are taking shape, and the green economic system The endogenous driving force of low-carbon circular development is growing stronger.

A new pattern of coordinated integration of climate governance and ecological and environmental protection is taking shape at an accelerated pace. The coordinated increase in pollution reduction and carbon reduction has become the overall approach to leading low-carbon development, coordinating pollutants and climate change The working concept of coordinated greenhouse gas emission reduction is constantly being improved. In the future, Guangdong will Be a practitioner, promoter and pioneer of good climate governance, strengthen the response to climate change and The unified planning, arrangement, implementation and supervision of ecological environment protection related work promotes the Beautiful Guangdong" will be completed as soon as possible.

The fourth quarter faces challenges

During the 14th Five-Year Plan period, the province's economic development will continue to drive the continued growth of energy demand.

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The low-carbon transformation of some industries and energy structures still faces certain difficulties, and there are still some outstanding issues in addressing climate change that need to be effectively addressed.

Our province's economic development and carbon emissions have not yet been completely decoupled. With the increase of population, urbanization and improvement of people's quality of life, the demand for energy is expected to continue to grow, which will add a certain amount of carbon reduction work in our province. pressure.

The clean level and utilization efficiency of energy consumption are not high enough. The level of industrialization continues to improve, but the proportion of coal consumption is higher than the world average, which is due to Germany, the United States and other countries It is 1.5 to 2 times that of developed countries. It is urgent to further tap the energy conservation and carbon reduction potential of the whole society and introduce Guide the green and low-carbon upgrading of industries.

The cross-departmental coordination mechanism is not yet sound. At present, the working mechanism for the relevant departments in our province to cooperate in responding to climate change is still unclear. Specific responsibilities are still unclear, and the applicability of statistical data related to energy and carbon emissions in various departments, There are still differences in uniformity, and the sharing channels of basic data have not yet been opened up.

The climate governance policy system still has shortcomings. Strategic planning guidelines for achieving the peak carbon neutrality goal need to be developed and improved. Modernization and transformation of management, low-carbon technological innovation, and grassroots supervision capacity building need to be strengthened. The incentive effects of economic policies such as finance, fiscal and taxation have not yet been fully exerted.

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Chapter II General Requirements

Section 1 Guiding Ideology

Guided by Xi Jinping Thought on Socialism with Chinese Characteristics for a New Era, we will thoroughly implement the Party's The spirit of the 19th National Congress and the 19th Plenary Sessions of the CPC Central Committee was fully implemented. Important speeches and important instructions and instructions, deeply practice Xi Jinping's ecological civilization thought, Unswervingly implement the new development concept, adhere to the coordinated promotion of carbon reduction, pollution reduction, green expansion and growth We should attach equal importance to mitigation and adaptation, and speed up the process of improving the level of knowledge, policy tools and means. measures, basic capabilities, and other shortcomings, and comprehensively strengthen the response to climate change and the ecological environment Coordinate and integrate relevant work on climate change protection, focus on improving governance capabilities to address climate change, and promote The province's green and low-carbon economic and social development continues to be at the forefront of the country and create new glory.

Section 2 Basic Principles

Systematic advancement and synergistic efficiency. Guided by the goals of carbon peak and carbon neutrality, we will reduce pollution and Carbon synergy is the overall starting point. We will strengthen top-level design and establish a sound overall plan to deal with climate change. Strengthen overall planning and systematization of policies, mechanisms and action systems for climate change and ecological environment protection Comprehensive layout, overall promotion, classified measures, and proper handling of development and emission reduction, overall and local, The relationship between the long-term and short-term, the government and the market, accelerating the comprehensive green transformation of economic and social development, and promoting coordinated progress in addressing climate change and high-quality economic development.

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Institutional innovation, trial and error. Actively explore innovative climate governance policies and management models that are in line with provincial conditions and have their own characteristics, and improve management systems and policy designs. Continue to actively Carry out demonstration pilot construction, vigorously promote green and low-carbon scientific and technological innovation, and accelerate the development of low-carbon technology Research and development and application demonstration to explore new paths for green and low-carbon development.

Double-wheel drive, two hands to better play the role of the government in addressing climate change give full play to the role of the market mechanism, and promote more effective government and effective market Actively guide all sectors of society to participate in the work of addressing climate change and stimulate the participation of the masses The enthusiasm, initiative and creativity in addressing climate change form effective incentive constraints mechanism.

Section 3 Main Objectives

By 2025, a high-quality development plan that is coordinated with the high-quality development of the economy and society and consistent with the ecological environment will be initially formed. A new approach to climate change response that is compatible with civilization construction and integrated with ecological environmental protection The green transformation of production and lifestyle has achieved remarkable results.

Industries and enterprises will reach peak carbon emissions first, further consolidating the foundation for carbon emissions peak. The institutional framework and market mechanism for neutralization have been basically established; the coordinated promotion of pollution reduction and carbon reduction The policy system and institutional mechanism for adapting to climate change have been basically formed.

The climate governance capacity has been significantly improved.

The total amount and intensity of carbon emissions have been effectively controlled. The province's carbon peak action has been fully launched. By 2025, carbon dioxide emissions per unit of regional GDP will drop by 20.5% compared with 2020, the carbon intensity level will continue to be at the forefront of the country, and the proportion of non-fossil energy consumption will reach about 32%.

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The energy consumption per unit of added value of industrial enterprises above designated size in the province has been declining year by year. The forest coverage rate in the province has reached 58.9%, the forest stock has reached 620 million cubic meters, and the ecological carbon sink has increased significantly. increasing.

Actively control non-CO2 greenhouse gas emissions. Methane emissions in the energy and waste sectors are effectively controlled, and nitrous oxide, hydrofluorocarbons, and perfluorocarbons in industrial production processes are The control of carbon dioxide and sulfur hexafluoride has been strengthened, the total amount of nitrous oxide emissions from farmland has been stable and declining, and livestock The control of greenhouse gas emissions from animal husbandry has been strengthened.

The ability to adapt to climate change has been comprehensively improved. Infrastructure and agriculture have obvious climate resilience The flood control and disaster reduction system of rivers, lakes and reservoirs has been further improved, and the flood and typhoon prevention capacity of coastal areas has been further improved. power has been significantly improved, ecosystem stability has been further improved, and biodiversity has been effectively Protection, climate disaster early warning and response capabilities continue to increase.

The climate governance capacity has been effectively enhanced. Pollution reduction and carbon reduction have been promoted in a coordinated manner, and climate governance and ecological A new pattern of coordinated integration of ecological environment protection has basically been formed. Improve greenhouse gas statistics, Accounting, monitoring and reporting systems to promote the "dual control" of energy consumption towards total carbon emissions and intensity "Dual control" transformation. Climate investment and financing mechanism is initially sound, and Guangdong carbon market construction continues The market carbon pricing system continues to improve.

The pilot system for carbon peak and carbon neutrality demonstration is sound and complete. Carbon pilot demonstration construction, improve and perfect supporting policies and evaluation indicator system, promote the park Actively explore carbon peak and carbon neutrality with enterprises, and build a number of carbon peak and carbon neutrality projects that can be replicated and promoted. Peak carbon neutrality demonstration cities (districts), parks and enterprises.

Looking forward to 2035, green production and lifestyle will be widely formed, and the province's carbon dioxide emissions will be

1. The target values of forest coverage rate and forest stock will be adjusted based on the integration results of forest, grassland and wetland data with the three national land surveys and the latest statistical caliber.

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After reaching the peak, the climate change will continue to decline, and the modernization of climate governance capacity will be basically achieved.

A solid foundation for carbon neutrality was laid years ago.

Table 1 Guangdong Province's 14th Five-Year Plan Climate Change Response Index System

First level indicator	Specific indicators	**Nature** of indicators for **2020-2025**		
Overall indicators	1. Reduced carbon dioxide emissions per unit of GDP ÿ%ÿ	[22.35]	20.5 Constraints	
	2. Energy consumption per unit of regional GDP decreased (%) [17.1] 3. Non-fossil energy accounted		14 Binding	
	for the proportion of primary energy consumption (%) 30.3 4. High-tech manufacturing industry		32 Anticipation	
Green transformation upgrade	added value accounted for the proportion of industrial enterprises above designated size Proportion of added value (%) 5.	31.1	33 Anticipation	
	Improvement of energy efficiency level of newly built residential buildings in urban areas (%) -		30 Anticipation	
	6. Improvement in energy efficiency of newly built public buildings in cities and towns		20 Anticipation	
	(%) - 7. Proportion of green buildings in cities and towns in newly built buildings (%) 63		100 Anticipation	
	8. Green buildings in cities with a permanent population of more than 3 million 9. Energy	70	80 Anticipation	
	consumption per unit of transport turnover of operating vehicles, CO2 10. Reuse rate of industrial water	.	4.0, 3.5 Anticipation	
	above designated size (%) 82.4 11. Recycling rate of urban domestic waste (%)		ÿ85Expectancy	
		.	35% Anticipation	
Ecological Protection and adapt to the atmosphere Climate Change	12. Forest coverage rate (%)	58.66	58.90 Binding	
	13. Forest stock (billion cubic meters)	5.84	6.2 Binding	
	14. Mainland natural coastline retention rate (%)	ÿ35	Country Reach the target	Constraints
	15. Average loss from meteorological disasters as a percentage of regional GDP Proportion(%)	0.19	ÿ0.15 Expected	
Demonstration pilot Construction	16. Number of low-carbon demonstration pilot projects	.	ÿ5 Anticipation	

Note: 1. The figures in [ ] are cumulative figures over the past five years.

2. The target values of forest coverage and forest volume will be integrated with forest, grassland and wetland data and the land survey data.

The results and the latest statistical caliber have been adjusted.

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Chapter III Main Tasks

Section 1 Strictly Control Greenhouse Gas Emissions

Solidly promote the province's carbon peak action and implement "dual control" of total carbon emissions and intensity system, promote high-quality green and low-carbon development of the economy and society, and effectively control non-CO Greenhouse gas emissions, increase ecosystem carbon sinks, adhere to the synergy of pollution reduction and carbon reduction, and increase Quickly form a green lifestyle.

1. Solidly promote the province's carbon peak action

Optimize the regional layout of green and low-carbon development. Establish and improve the "three lines and one list" ecological The environmental zoning control system is constructed to adapt to the development pattern of "one core, one belt and one region". Strengthen the orientation and task requirements of green and low-carbon development, and continuously optimize major infrastructure Infrastructure, major productive forces and public resource layout, and build a carbon peak and carbon neutrality A new pattern of land space development and protection. Seize the Guangdong-Hong Kong-Macao Greater Bay Area, Shenzhen and China's characteristics The construction of the “dual zones” of the socialist pilot demonstration zone and the two cooperation zones of Hengqin and Qianhai is of great importance. Great opportunity, promote the Pearl River Delta region to take the lead in achieving comprehensive green transformation, and work together with Hong Kong and Macao to build Green and low-carbon bay area, exploring the Guangdong-Hong Kong-Macao Greater Bay Area and Shenzhen's pioneering efforts in socialism with Chinese characteristics Demonstration area of green and low-carbon development of new path. Promote unified planning, scientific deployment of Guangdong, Hong Kong and Macao The construction of urban green space in the Bay Area will increase urban wetlands, green spaces and water bodies to alleviate the heat island effect. Optimize the networked spatial pattern of efficient connections between cities and create a vibrant world-class climate-adaptive urban cluster. Promote the optimization of industrial layout in the east and west wings and dig deep into the

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Energy conservation and emission reduction potential, accelerate the creation of a world-class coastal heavy industry belt and a national marine economic development demonstration zone. Promote the northern ecological development zone to adhere to ecological priority and green development. Strictly control development intensity and promote industrial ecology and ecological industrialization.

Establish a system that focuses on carbon emission intensity control and is supplemented by total carbon emission control. Fully implement the implementation opinions of Guangdong Province on carbon peak and carbon neutrality and the implementation plan for carbon peak before 2030. Plan, organize and promote major reforms, major demonstrations and major projects in carbon peak and carbon neutrality. Comprehensively consider the economic and social development level, carbon emission level, main functional positioning, Factors such as regional development pattern, according to the "14th Five-Year Plan" carbon emission reduction plan issued by the state to Guangdong Province The carbon emission intensity reduction requirements were broken down and implemented in various cities. Study the establishment of a total carbon emission control system. Encourage the Guangdong-Hong Kong-Macao Greater Bay Area city cluster to take the lead in trial implementation. Actively explore low-carbon development models and supporting policies for “dual control” of total carbon emissions and intensity Mechanism to promote the transition from "double control" of energy consumption to total carbon emissions as soon as possible across the province Create experience in the dual control of carbon emissions and intensity. Conduct research on the carbon budget system and carry out pilot projects.

Support qualified cities, key industries and enterprises to reach the peak first. To guide all cities and municipalities to speed up the formulation of carbon The implementation plan for the peak period should be implemented, regional differences and development rhythms should be grasped, and reasonable peak targets should be set. Encourage cities and prefectures with conditions to reach the peak first. Focus on key areas and key links of carbon emissions, organize Organize and implement the "15th Carbon Peak Action Plan". Guide key industries and enterprises with conditions to formulate We will implement the carbon peak path measures to accelerate the green and low-carbon transformation of industry and high-quality development. Improve the overall energy efficiency of industry and promote key industries such as steel, cement, and ceramics in the "Ten We will be the first to achieve peak demand during the "Fourth Five-Year Plan" period and promote the overall peak-reaching process in the industrial sector.

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2. Promoting high-quality and low-carbon development of industries

Promote the transformation and upgrading of industrial structure. Accelerate the green development of agriculture and promote Agricultural intensive production technology that is conducive to carbon peak and carbon neutrality, promotes input reduction, Clean production, waste resource utilization, and ecological industrial model will achieve agricultural carbon sequestration and efficiency improvement. Deeply implement the "six major projects" for high-quality development of the manufacturing industry, consolidate and improve the new generation Ten strategic pillar industry clusters such as electronic information, forward-looking layout of semiconductors and integrated circuits Ten strategic emerging industry clusters such as roads are planned to develop future industries such as artificial intelligence. Promote the leap of manufacturing industry from agglomeration development to cluster development, and promote the advanced development of industrial foundation and industrial chain modernization, and promote the integration of digitalization, intelligence and greening of traditional industries Development. Accelerate the cultivation of new low-carbon, zero-carbon and negative-carbon business forms, new models and new industries, and promote the development of marine The industrialization of carbon dioxide storage in geological structures will promote the Internet, big data and other new technologies. Deeply integrate emerging technologies with green and low-carbon industries. Promote green development in key industries and key areas. We will continue to build a number of green factories, green design products, green parks, Green supply chain, and gradually build a green manufacturing system covering the entire industrial chain and the entire life cycle of products. Accelerate the extension of productive services to specialization and high-end value chains, and life services The industry is upgrading to high quality and diversification, and the service industry is accelerating green transformation and cooperating with advanced manufacturing. Deeply integrate manufacturing and modern agriculture to improve the low-carbon development level of the service industry. During this period, the energy consumption per unit of added value and carbon emissions of industrial enterprises above designated size in the province decreased year by year. By 2025, the added value of high-tech manufacturing will account for the added value of industries above designated size. Reached 33%.

Resolutely curb the blind development of high-energy-consuming and high-emission projects. Strictly implement the Energy Conservation Law, Environmental Protection Law and other laws and regulations, as well as industrial policies such as the "Guidelines for Industrial Structure Adjustment"

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Improve the policy and planning layout, strictly control the scale of high energy consumption and high emission (hereinafter referred to as "two high") industries.

Improve the access conditions for high energy consumption, high pollution and resource-based industries, and the provincial and municipal "three lines and one

When updating and adjusting the "two high" project access and control requirements, the Pearl River Delta region

The district has formulated and issued a stricter "Guide Catalogue for Industrial Structure Adjustment", prohibiting the construction and

expansion of cement, flat glass, chemical pulping, leather and leather, and steel outside the national plan.

Iron and crude oil processing projects. Continue to reduce the proportion of high energy-consuming industries in the overall manufacturing industry

The two highs are now being upgraded.

The energy efficiency level of the project must reach the domestic advanced level, except for the coal-fired power projects planned and laid out by the state.

In addition, new “two high” projects involving coal consumption shall implement coal consumption reduction and substitution.

The scale needs to come from within the province.

Accelerate the low-carbon transformation of traditional industries such as steel, petrochemicals, cement, ceramics, and papermaking

Promote the elimination of obsolete steel production capacity in accordance with laws and regulations, and carry out the "return to production capacity" of steel

Promote the integrated development of scrap steel recycling, dismantling, processing, classification and distribution,

Improve the recycling level of scrap steel resources, orderly guide the development of short-process electric furnace steelmaking, and promote

Carry out integrated layout of steel, coking and sintering, actively promote the integration of steel, building materials, petrochemicals

Coupling development of industries such as engineering, electricity, and nonferrous metals. Explore the development of non-blast furnace ironmaking and hydrogen energy metallurgy

Pilot demonstration of low-carbon metallurgical technologies such as refining, carbon capture and storage, etc.

The province's steel rolling industry has basically completed the replacement of clean energy.

Strengthen the control of carbon dioxide emissions, accelerate the promotion of reducing oil and increasing chemical production, and actively develop green hydrogen chemical production

Industry. Accelerate the adjustment of petrochemical fuel and raw material structure, encourage coal substitution, and promote petrochemical

Lightweight chemical raw materials, promote the application of raw material optimization, energy cascade utilization, material recycling, process

reengineering and other process technologies and equipment, promote the development of petrochemical industry, metallurgy, construction

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Coordinated development of industries such as materials and chemical fibers, and actively promote the supply-side structural reform of the petrochemical industry to achieve scale, intelligence, greening and integrated refining.

The transformation of cement kilns to chemical, product and low-carbon is encouraged to coordinate the disposal of sludge, domestic waste, Industrial waste residues and other wastes. Rationally control carbon emissions in the cement production process, explore pilot demonstrations of carbon capture and storage of cement kiln tail gas. Promote the normalization of staggered cement production and promote Apply the second generation of new dry process cement technology and equipment. Promote the application of electric kilns in the ceramic industry Energy-saving and low-carbon technologies such as coal-fired furnaces and spray towers and dry pulverizing.

The industry promotes the concentration of small and medium-sized enterprises in the park, implements unified power supply and heating, and promotes energy conservation process technology, promote forest-pulp-paper integration, and enhance the intensive, high-end, Green level. On the basis of ensuring the stable supply of natural gas, steadily promote the development of ceramics, "Replacing coal with gas" in the papermaking industry.

3. Build a clean, low-carbon, safe and efficient energy system

Vigorously develop non-fossil energy. Under the premise of ensuring safety, efficiently build Huizhou Taipingling Nuclear Power Phase I Project, actively promoted the start of Lufeng Nuclear Power, Lianjiang Nuclear Power and other projects, It will also promote the preliminary work of a number of subsequent projects and do a good job in protecting nuclear power plant sites.

Develop offshore wind power and build a 10 million kilowatt offshore wind power base in eastern and western Guangdong. Centralized onshore wind power should be developed cautiously in areas with relatively rich resources and far away from residential areas.

Develop distributed onshore wind power according to local conditions. Vigorously increase the scale of photovoltaic power generation and adhere to the centralized Centralized and distributed development are carried out simultaneously, centralized photovoltaic power station projects are built according to local conditions, and Encourage the development of rooftop distributed photovoltaic power generation, and explore the development of county-wide rooftop distributed photovoltaic power generation Develop pilot demonstration projects and actively promote the construction of photovoltaic building integration. Develop and utilize biomass energy according to local conditions. Actively promote the transmission of clean electricity from outside the province such as hydropower in southeast Tibet to Guangdong.

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Build a new power system with new energy as the main body, accelerate the cultivation of hydrogen energy, energy storage, smart energy, etc., and vigorously promote the construction of "new energy + energy storage" projects. By 2025, non-chemical The proportion of fossil energy installed capacity (including Xidian) reached about 49%. During the 14th Five-Year Plan period, new The installed capacity of wind power, photovoltaic power and nuclear power generation will be increased by approximately 20 million kilowatts, 20 million kilowatts and 2.4 million kilowatts respectively. The scale of electricity input from the west will be increased to 45 million kilowatts. Pumped storage power will be added. The installed capacity of the station is 2.4 million kilowatts, and the proportion of non-fossil energy in primary energy consumption reaches About 32%.

Promote clean and efficient use of fossil energy. Strictly and reasonably control the growth of coal consumption. Orderly shut down coal-fired power units with small scale, high coal consumption, long service time and high emission intensity, Gradually reduce the proportion of coal-fired power. The Pearl River Delta region prohibits the construction and expansion of coal-fired and oil-fired power plants. The coal-fired and oil-fired power stations of the group or enterprise are allowed to replace and reconstruct high-voltage power stations according to national policies. Promote energy-saving and consumption-reducing reforms of existing coal-fired power units The three reforms of building, heating and flexibility transformation are linked together, and key areas, The coal-fired power plants and coal-fired boilers in key industries will be converted from coal to gas, and the “coal-fired power plants” and “coal-fired boilers” will be converted from coal to gas. Actively carry out the shutdown and integration of coal-fired boilers to fully release the cogeneration, The heating potential of industrial waste heat and other sources will be fully utilized, and clean energy substitution for industrial furnaces and boilers will be implemented in an orderly manner. Rationally plan the layout and construction of peak-shaving gas-fired power plants in load centers such as the Pearl River Delta, and actively promote industrial (production) Accelerate the construction of natural gas cogeneration and distributed energy projects in industrial parks. The main pipeline network "County-to-County Project" has basically realized the natural gas trunk except for Nan'ao County The pipeline network is connected to all counties, and natural gas is comprehensively promoted in urban residential gas, industrial fuel, Efficient use in transportation, commerce and other fields. Strengthen the quality control of oil products in the transportation field, vigorously curb the excessive growth of oil consumption, and keep oil consumption within a reasonable range.

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By 2025, the proportion of natural gas in primary energy consumption in the province will reach 14%, and the installed capacity of gas-fired power generation will reach about 64 million kilowatts.

kilowatt.

Comprehensively improve energy efficiency. Adhere to the principle of giving priority to conservation and efficiency, establish an energy budget management system, strictly review project energy conservation, and do a good job in energy conservation review and energy consumption dual control. To ensure the connection of carbon emission control, new projects must comply with national industrial policies and achieve energy efficiency The industry is advanced.

Focusing on industries such as paper and textile printing and dyeing, we will conduct in-depth energy conservation and emission reduction diagnosis and establish energy efficiency Advanced and backward lists, comprehensively promote energy-saving transformation and upgrading, and improve production processes and technologies Improve the green level of equipment, aim at the international advanced level, and carry out energy efficiency benchmarking activities. Promote no less than 50 energy efficiency leaders in the industry with leading energy efficiency.

Promote the energy efficiency of new infrastructure, optimize the construction layout of data centers, and promote the Green upgrading and transformation of the data center. Focus on industrial parks with high concentration of "two high" projects points, promote the overall optimization of the energy system and the cascade utilization of energy. Continue to promote public institutions Existing building maintenance structures, refrigeration, lighting, elevators and other comprehensive energy systems and facilities Equipment energy-saving transformation, enhance the demonstration and driving role. By 2025, the province's units and regions will The energy consumption of GDP decreased by 14% compared with 2020.

The proportion of production capacity of key industries such as calcium carbide and data centers reaching energy efficiency benchmark levels exceeds 30%, flat glass over 20%, synthetic ammonia over 15%.

4. Accelerate the promotion of green and low-carbon development in urban and rural areas

Promote the green and low-carbon transformation of urban and rural construction. Promote the development of urban clusters, improve the building demolition management system, put an end to large-scale demolition and construction, and rely on green towns, green communities and beautiful

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Green rural construction, establish and improve a green and low-carbon urban and rural construction management mechanism, and promote the green and low-carbon transformation of urban and rural construction and management models.

Standards, implement green building full-process construction management and a unified green building identification system.

Vigorously promote prefabricated buildings. Government-invested projects should take the lead in developing prefabricated buildings.

In principle, affordable housing and large public construction projects should implement prefabricated buildings.

The green design concept of adapting to local conditions, low-carbon development and cultural heritage forms a climate-friendly

Green building design technology that includes environment, space control, construction methods, energy utilization, etc.

Implement green building performance improvement plans, inherit, promote and innovate Lingnan characteristics

Green building technology that is suitable for subtropical climate.

By 2025, the proportion of star-rated green buildings will reach more than 30%, and new large-scale

All public buildings are of star-rated or above.

Prefectural-level cities in the east, west and north of Guangdong with a built-up area ratio of more than 35% and a permanent population of more than 3 million

The rate in the central urban area reached over 30%, and in other areas it reached over 20%.

Improve the energy efficiency of buildings. Vigorously develop ultra-low energy consumption buildings and carry out near-zero energy consumption

Building, zero-carbon building pilot demonstration. Inherit and innovate the application of passive buildings with Lingnan characteristics

energy-saving technologies for modern buildings, and gradually improve the mandatory energy-saving standards for newly built green buildings.

Improve the energy efficiency of buildings. Combined with urban renewal actions, renovation of old urban communities, and rural

The renovation of dilapidated houses in villages promotes energy-saving and green transformation of existing buildings, and vigorously promotes contract energy management

Establish a data sharing mechanism for urban building electricity and gas consumption

Strengthen the intelligent monitoring and fine management of energy consumption in commercial and public buildings, and improve

Building energy consumption monitoring capabilities. Clarify the energy consumption limit standards for public buildings and implement a public

building energy consumption over-limit notification system. Vigorously promote the application of renewable energy buildings and improve urban and rural

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The level of electrification of residents' lives will reach 500 million by 2025, and the replacement rate of renewable energy in urban buildings will reach 100 million by 2025.

8%ÿ

5. Promote the construction of a green, low-carbon integrated transportation system

Accelerate the adjustment of transportation structure. Build a high-capacity, fast, low-carbon passenger transportation service system with high-speed railways, intercity railways, and urban rail transit as the main body and multiple networks integrated. We will speed up the transition from road to rail and from road to water for bulk cargo transportation. Public transportation infrastructure construction such as transportation, bus lanes, and bus rapid transit systems Build a slow traffic network. Relying on large hubs such as ports, airports, and railways, Strengthen the improvement and function enhancement of intermodal transport facilities such as road-rail, rail-water, and air-rail. Promote the integrated development of "railway + port + logistics" functions. Green freight delivery demonstration projects will be built in cities such as Foshan. By 2025, container railways will Water transport has an average annual growth rate of more than 15%, and the city green The proportion of color travel has reached 80%.

Accelerate the promotion and application of new energy transportation tools. Increase the use of clean energy and new energy Application in transportation equipment, promoting hydrogen energy buses, hydrogen energy rail transit and electric By 2025, the number of newly added or updated public All vehicles delivered will be electric and hydrogen fuel cell vehicles. All cruising taxis (including online taxis) use new energy vehicles. The proportion of new additions (including renewal) in the district shall not be less than 80% and shall increase by 5 percentage points (including Energy-saving vehicles). New energy vehicles (including energy-saving vehicles) among the official vehicles newly added and updated each year The proportion shall not be less than 60%. The proportion of new or updated urban logistics distribution, light postal express, and light sanitation vehicles using new energy vehicles in cities above the prefecture level shall reach more than 80%.

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Rapidly phase out old vehicles and construction machinery with high energy consumption and high emissions. Accelerate the layout of urban and rural public charging and swapping networks, and achieve full coverage of fast charging stations in highway service areas by 2025. By 2025, the number of new energy vehicles in the province will reach 1 million, and new energy vehicles will The proportion of new car sales reached more than 20%, of which Guangzhou and Shenzhen reached about 40%. Accelerate the application of LNG-powered ships and the layout and construction of inland LNG filling stations, and Build an inland river ship LNG application demonstration project and put the refueling station into operation by the end of 2022 6, forming an application scale of 300 LNG ships. Promote the construction of port shore power, Encourage ships to use shore power when docked, and give priority to its use in port loading and unloading machinery and transport equipment Electricity or natural gas. Implement the management policy of the ship emission control area in the Pearl River Delta waters. Actively Promote the use of renewable energy such as solar energy and wind energy in the construction and operation of transportation infrastructure Encourage driving training companies to add new energy small cars and guide the existing old fuel Driving training small cars are being eliminated at an accelerated pace.

6. Controlling non-CO2 greenhouse gas emissions

Control methane emissions from oil and gas systems. Strengthen offshore oil and gas exploration and development, and natural gas Methane recovery in industrial processing and transportation, crude oil transportation, oil refining, oil and gas consumption, etc. Utilize, coordinate and control volatile organic compound emissions, and promote the research and development of low-cost coordinated control technology Development and application will improve the level of methane emission monitoring and perfect the statistical mechanism of methane emission accounting. Strengthen the inspection and maintenance of oil and gas pipelines and gas storage facilities to reduce the daily By 2025, the oil and gas industry will reduce its methane emission intensity per unit of oil and gas equivalent to The oil and gas exhaust gas recovery rate can reach more than 50%.

Control greenhouse gas emissions from industrial production processes. Reduce non-CO2 greenhouse gas emissions from industrial production processes through process technology improvements and end-of-pipe treatment.

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Control the emission of by-product trifluoromethane (HFC-23), promote enterprises to establish operation records of HFC-23 by-product facilities and destruction and disposal facilities, monitor and Metering and regular inspection of HFC-23 leakage and emission. Establish and improve the power system Unify the sulfur hexafluoride control policy and further strengthen the recovery and utilization of sulfur hexafluoride. Actively promote the research and development and application of alternative technologies for hydrofluorocarbons and sulfur hexafluoride, and promote the use of warming Production and use of low potential alternatives to hydrofluorocarbon refrigerants.

Control nitrous oxide emissions during the production process and promote nitrous oxide terminal treatment technology.

Control methane and nitrous oxide emissions in the agricultural sector. Promote water conservation and energy conservation in livestock and poultry farming. and other technologies such as dry manure removal and microbial fermentation, and carry out demonstration projects on livestock and poultry breeding standardization. Create a model to control greenhouse gas emissions from livestock and poultry farming. Deepen the implementation of fertilizer and pesticide reduction Improve efficiency, further promote soil testing and formula fertilization, integrated prevention and control of major crop diseases and insect pests, and green Accelerate the research and demonstration of comprehensive utilization technology of straw, and promote the By 2025, the comprehensive utilization rate of livestock and poultry manure in the province will reach more than 80%, mainly The utilization rate of fertilizers and pesticides for crops should be kept above 43%, and the comprehensive utilization rate of straw should be kept above 43%. Set at above 86%.

Control methane and nitrous oxide emissions from waste treatment.

The construction of a "waste-free bay area" will further increase the construction of waste harmless treatment facilities. Promote the harmless treatment and resource utilization of kitchen waste, and encourage the harmless treatment of residues. Make fertilizers, increase the proportion of domestic waste incineration capacity, and reduce landfill disposal volume. Strengthen methane control and recycling in wastewater treatment and landfill. The recycling rate of urban domestic waste in the province reached more than 35%, and urban domestic waste was rendered harmless. The processing rate is no less than 99%.

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7. Increasing the amount of carbon sinks in natural ecosystems

Cultivate and enhance the carbon sequestration capacity of ecosystems. Fully implement the Green Guangdong Action. Improve the natural forest protection system, strengthen forest cultivation and transformation, and promote the quality and efficiency of public welfare forests. Carry out key ecological projects such as high-quality water source forest projects and large-diameter timber cultivation. Improve the carbon sink capacity of the ocean and enhance soil carbon sink. By 2025, the forest coverage rate of the province will reach 2 The forest stock volume reached 620 million cubic meters, and 5,500 hectares of mangroves were planted. A moment.

Strengthen the basic support for ecosystem carbon sinks. Establish an ecosystem carbon sink monitoring and accounting system. system, conduct carbon sink background surveys in forests, wetlands, oceans, soils, karsts, etc., and conduct scientific Assess the carbon budget of ecosystems. Take Shenzhen and Zhanjiang as marine pilot cities and carry out Monitoring carbon sinks in mangrove and seagrass ecosystems, estimating ecosystem carbon storage, and studying Establish typical coastal ecosystem monitoring and assessment methods and carbon sink monitoring technology systems. Shaoguan is a pilot city to carry out forest ecosystem carbon sink monitoring, Conduct biomass surveys and monitoring of soil and litter, and estimate ecosystem carbon storage. Improve the forestry ecological protection compensation mechanism that can reflect the value of carbon sinks and improve forestry carbon sinks Trading market mechanism.

8. Promote synergistic efficiency in pollution reduction and carbon reduction

Incorporate the carbon peak and carbon neutrality requirements into the "three lines and one list" three-zone management and control system. In line with the requirements of national land space planning zoning and use control, strengthen spatial guidance and zoning measures, Enhance the guiding role of regional environmental quality improvement goals on energy and industrial layout, and study

2 The target values of forest coverage and forest stock will be adjusted based on the integration of forest, grassland and wetland data with the data of the "three national land surveys" and the latest statistical caliber. 3 Including ecological protection red line, environmental quality bottom line, resource utilization upper line and ecological environment access list.

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Establish an industry entry and exit list system guided by regional environmental quality improvement and carbon peak targets. Support Zengcheng District of Guangzhou City to explore the "three lines and one list" ecological environment as a pilot

Zoning control promotes the technical path and management model of coordinated control of pollution reduction and carbon reduction, and encourages

Actively explore distinctive coordinated paths for pollution reduction and carbon reduction and the optimization model of "three lines and one list" ecological

environment zoning control. Increase efforts to adjust the structure and optimize the layout of severely polluted areas.

To accelerate the withdrawal of backward and excess production capacity in the Pearl River Delta region.

Promote coordinated control in the field of air pollution prevention and control. Optimize the technical route of governance and increase

Volatile organic compounds (VOCs), a common precursor of fine particulate matter (PM2.5) and ozone pollution

and nitrogen oxide emission reduction efforts, and coordinate air pollution control and greenhouse gas emission control.

In the treatment of VOCs and other air pollutants, source substitution measures should be given priority, and research and development and promotion of VOCs

Resource recycling and low-energy waste gas treatment technologies such as low-temperature catalysis promote the

Energy conservation and consumption reduction in pollution control facilities. Promote in-depth treatment and energy conservation in key industries

Carbon reduction actions can be taken to explore the coordinated control of atmospheric pollutants and greenhouse gas emissions

Improve engineering pilot projects. Encourage cities with conditions to take the lead in building carbon dioxide peak and air

Demonstration of collaborative governance to achieve quality improvement targets. Strengthen the management of ozone-depleting substances and hydrofluorocarbons

Carbon management, accelerate the transformation of production lines using hydrochlorofluorocarbons, and gradually phase out hydrochlorofluorocarbons.

Promote the coordinated control of mobile source air pollutant emissions and carbon emissions.

Promote coordinated control in the field of water environment governance. Promote energy conservation and consumption reduction in sewage treatment plants.

Optimize industrial processes, improve treatment efficiency, and explore the transformation of traditional sewage treatment from simple pollution

Develop urban sewage treatment and carbon resource utilization

Emissions calculation, optimize the energy consumption and carbon emission management of sewage treatment facilities. Strengthen the use of recycled

water, reduce industrial water intake, and use recycled water, rainwater storage, seawater desalination, etc.

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By 2025, the reuse rate of industrial water above designated size in the province will be raised to

Over 85%.

Promote coordinated control in the field of soil pollution control. Strictly manage the redevelopment and utilization of polluted construction land, rationally plan the use of polluted land, and encourage industries such as chemical industry to Moderately and severely polluted plots are planned to expand ecological space and reduce energy consumption for restoration. Low-carbon restoration, optimizing soil pollution risk management and restoration technology routes, focusing on energy conservation Reduce consumption.

Promote coordinated control in the field of resource recycling. Promote the construction of "zero-waste cities" Give full play to the synergistic effect of pollution reduction and carbon reduction. Build an industrial circulation system in the park to promote The industrial cycle coupling within the park improves the production, consumption and flow of green and low-carbon circular development. Encourage coal-fired power plants, cement kilns, etc. to coordinate the disposal of sludge, promote the Classify and treat solid waste such as industrial waste, and strengthen the recycling of renewable resources. Industrial solid waste such as fly ash, smelting slag, and coal gangue can be used as raw materials for building materials production. Recycling of waste paper, waste plastic, waste tires, waste metal, waste glass and other renewable resources. Promote the recycling of new types of waste such as retired power batteries, photovoltaic modules, and wind turbine blades Reduce the landfill of organic waste and strengthen the utilization of leachate and Coordinated control of odor and greenhouse gases, and promotion of landfill gas collection and utilization facilities Construction. Steadily promote the diversified development and utilization of biomass energy according to local conditions. Prohibit persistent Illegal production of organic pollutants and mercury-added products, reducing toxic and harmful chemicals from the source By 2025, the annual utilization of bulk solid waste in the province will reach 55 million tons, the comprehensive utilization rate of new bulk solid waste will reach 60%, and the existing bulk solid waste will reach 100 million tons.

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Waste will be reduced in an orderly manner, and the resource utilization rate of urban domestic waste will be no less than 60%.

9. Accelerate the formation of a green and low-carbon lifestyle

Advocate simple and moderate green consumption. Increase the government's green procurement efforts and expand green Product procurement scope: implement the preferential procurement and mandatory government procurement policies for environmental label products, energy-saving and water-saving products, and Guangdong Province carbon label products, and gradually expand the green procurement system To state-owned enterprises. Actively play the leading role of green consumption and vigorously promote energy conservation, environmental protection and low Carbon products, improve energy-saving home appliances, high-efficiency lighting products, water-saving appliances, green building materials, etc. Green products and new energy vehicles promotion mechanism, encourage cities and prefectures with conditions to promote public Provide appropriate support for green consumption. Practice economy and oppose waste and excessive consumption. In order to actively practice the "Clean Plate Campaign".

Actively promote the creation of green life. Implement the "Overall Action Plan for Creating Green Lifestyle", Encourage the development of energy-saving offices, green families, green schools, green communities, and green transportation. Promote green office, green residence, Green travel, reducing the use of disposable products and other low-carbon behaviors, actively advocating intensive and economical Actively promote the household waste classification mechanism and guide the public to Participate in the classification of domestic waste, support voluntary tree planting, and prohibit open-air burning. Burning garbage and straw.

Strengthen the content construction of publicity and education on climate change response. Establish and improve green and low-carbon living standards. Active publicity and promotion system, make full use of various media platforms, give full play to the Internet, Weibo, WeChat public accounts and other new media have played a role in widely conducting public opinion propaganda and guidance, effectively raising the level of Raise public awareness of green and low-carbon development. Incorporate green and low-carbon concepts into the education system, and use education to drive the whole society to practice green and low-carbon development. Encourage social organizations to carry out green and low-carbon public welfare activities and invest

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Green and low-carbon public service advertisements promote the dissemination and popularization of low-carbon concepts, and widely publicize and promote simple, moderate, green and low-carbon life concepts and lifestyles.

Section 2 Actively Adapting to Climate Change

Coordinate efforts to adapt to climate change and protect and restore ecosystems, and enhance natural ecosystems Improve climate resilience in key strategic economic regions and ecological security areas strengthen the sustainability of socio-economic development, and enhance monitoring of extreme climate events. Build a disaster prevention and mitigation system to improve the assessment of adverse impacts and risks of climate change level, and the climate change adaptation actions in the strategic region of the Guangdong-Hong Kong-Macao Greater Bay Area have been effectively carried out.

1. Enhancing climate resilience in the natural ecological sector

Optimize water resource management and conservation. Strengthen water conservation in major rivers such as the Dongjiang River, Xijiang River, and Beijiang River. The connection between water source supply areas will build a multi-source water supply network in the city. Pearl River Delta water resources allocation, Hanjiang, Rongjiang and Lianjiang river system connection, Guangzhou Beijiang diversion Water and other engineering construction, and promote the full start of construction of the Guangdong Water Resources Allocation Project around the Beibu Gulf. We will continue to increase our efforts in water supply infrastructure construction and water resource protection in Hong Kong and Macao, and ensure the water supply between Guangdong and Hong Kong will be safe. High-quality and safe water supply in the Greater Bay Area. Deeply implement the national water conservation action and strengthen water resources security. We will impose constraints on water consumption, implement dual controls on both total water consumption and intensity, and promote the construction of a water-saving society. By 2025, a rigid water resource constraint system will be basically established, and water resource utilization efficiency and The benefits have been greatly improved. Strengthen urban water conservation and loss reduction, and implement water conservation in national land space planning, All aspects of urban construction and management. Strengthen water-saving transformation of backbone canals in agricultural irrigation areas, supporting field projects, management of flood-prone areas and agricultural water use management to achieve the whole process of water delivery and water use.

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Water conservation. By 2025, the compliance rate of water-saving society construction in county-level administrative areas will reach 60%, and the effective utilization coefficient of farmland irrigation water will reach 0.535.

Coordinate the integrated protection and restoration of mountains, rivers, forests, farmlands, lakes and grasslands. Actively promote the restoration of degraded natural forests Restoration, and overall promotion of key forestry ecological projects and ecological restoration of damaged abandoned land. Effectively protect forest resources and strengthen monitoring of forest disasters such as fires and harmful biological invasions Strengthen prevention and control, improve ecological security and green development capabilities, and protect the integrity of the ecosystem. Strengthen the construction of urban park green space, suburban ecological green space, green space Improve the urban green space system. Strengthen the protection of wetland systems and vigorously promote Wetland park construction will comprehensively enhance the ecological service functions of river and lake wetlands. By 2025, The province's natural forest reserves remain stable at over 2.4 million hectares.

Stabilize and improve the marine and coastal ecosystems. Adhere to the principles of ecological use of the sea and intensive use of The principle of the sea is to implement the control requirements of marine ecological space and development and utilization space, strictly guard the sea Strengthen the protection of natural coastlines and marine biodiversity and alien species control, further reduce pollutants entering the sea from rivers, and enhance the marine ecosystem Improve the quality, stability and climate resilience of the coastal marine ecosystem. Implement coastal wetland restoration, ecological seawalls, natural coastlines, mangroves, seagrass beds, etc. Typical marine ecosystem restoration projects, including coastal shelterbelt basic forest belts and deep shelterbelts Forest protection construction, complete the restoration of 2,500 hectares of mangroves by 2025.

2. Improving the ability of key regions to respond to climate change

Enhance the ability of the strategic region of the Guangdong-Hong Kong-Macao Greater Bay Area to adapt to climate change. Guangdong, Hong Kong and Macao ecological corridors and biodiversity protection networks, strengthen the use control of natural ecological space, and carry out cross-border joint protection of coastal wetlands.

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Improve the ability of the economic belt to resist, recover and control typhoons, storm surges, rainstorms and floods. Promote the establishment of an interactive mechanism among relevant departments in Guangdong, Hong Kong and Macao, and gradually improve the

The three places have established a mechanism for disaster consultation, information exchange and coordinated response, with a focus on strengthening the coordination of sea level rise.

The coordinated monitoring, early warning and emergency response of typhoons, storms and marine disasters.

Sharing and business cooperation, improving the comprehensive disaster prevention and mitigation system such as flood and typhoon prevention. Strengthening seawalls

Strengthen the construction of key projects such as the reinforcement of the main and tributary rivers of the Pearl River, and strengthen the Greater Bay Area

Coordinated construction of urban ecological environment integrating land and sea, building coastal ecological barriers,

By 2025, the seawall compliance rate in the province will reach 80%.

Focus on strengthening the Nanling Mountain Forest and Biodiversity Ecological Functional Zone in northern Guangdong

Part of the ecological protection, promote the construction of Nanling National Forest Park, build and consolidate the northern

Ecological barrier. Vigorously promote the construction of water conservation forests and soil and water conservation forests, strengthen soil and water conservation

Strengthen the protection of ecological corridors of important rivers and water systems in the northern ecological barrier region.

Protection.

3. Strengthening the ability of the economic and social sectors to adapt to climate change

Strengthen climate resilience in the agricultural sector. Ensure food security and supply and demand of important agricultural products

Optimize the agricultural development layout and agricultural function positioning in the province, coordinate and promote agricultural water conservation,

Fertilizer-saving, pesticide-saving, land-saving and energy-saving technologies should be promoted and applied according to local conditions, such as dryland agriculture,

Various agricultural adaptive technologies such as drought conservation, cultivating and selecting high temperature resistance, cold resistance, drought resistance,

Crop varieties with strong adaptability such as flood resistance will be developed to accelerate the development of modern agriculture.

We should take advantage of the warm winter climate and fallow land resources in the north and southwest of China to expand winter crop production.

We should give priority to ecological development in agricultural development and strengthen soil and water conservation and ecological protection.

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Invasive biological control and protection of agricultural biodiversity.

Improve urban climate adaptation and protection capabilities. Give full consideration to urban and rural construction planning. Climate carrying capacity and climate change impacts, and make overall arrangements for urban construction, industrial development, and production. Ecological conservation, infrastructure and public services, rationally plan urban layout and functions. Actively respond to the heat island effect and urban waterlogging, scientifically formulate rainstorm intensity formulas and determine rainstorm intensity. Accelerate the construction of sponge cities and resilient cities, appropriately improve urban flood control and management Waterlogging standards should be met, and urban river networks should be preserved and gradually restored. The construction of public places such as the

By 2025, the compliance rate of major river embankments in the province will be

Reach 85%, the flood (tide) prevention capacity of the central areas of Guangzhou and Shenzhen shall not be less than 200

The flood (tide) prevention capacity of the urban center of other prefecture-level cities shall not be less than that of a 100-year flood.

Promote the construction of new urban infrastructure and ensure the systematic, intelligent and

Green construction and safe and stable operation.

Accelerate the development of climate change environmental health and public health. Improve climate change Strengthen the prevention and control mechanisms for the impacts of climate change on vulnerable areas and vulnerable populations. Strengthen real-time high temperature health monitoring and target different thresholds for different groups of people. Issue high temperature warning information, strengthen monitoring and early warning of climate-sensitive disease transmission risks, and plan Health Action. Improve the prevention and control mechanism of the impact of climate change on human health, encourage the formulation of and improve public health emergency response plans for extreme weather events and climate crises, and enhance Emergency protection skills for all people to cope with extreme climate events.

4. Strengthening capacity building for climate disaster prevention and mitigation

Strengthen the precise monitoring and accurate prediction and early warning of climate disasters.

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Comprehensive ecological meteorological observations will be carried out in ecologically sensitive and vulnerable areas to enhance the meteorological monitoring capabilities of forests, wetlands, rivers and reservoirs, offshore coasts, farmlands, and towns.

Improve the ability to identify risks early and accurately forecast them.

The automatic observation and early warning system for drought, tropical cyclones and high-impact climate events will be improved to realize the comprehensive monitoring of extreme climate events and disasters with high temporal and spatial resolution.

By 2025, the effective early warning of sudden disaster weather will reach 60 minutes in advance.

More than one minute.

Improve the climate disaster response and prevention system and emergency response mechanism.

Disaster safety risk prevention and mitigation mechanism, strengthen the comprehensive disaster risk management under the influence of climate change

Build monitoring, early warning and assessment systems. Establish a safety risk investigation and safety warning system for climate disasters.

Strengthen the construction of emergency public infrastructure such as emergency communications and fire protection, and strengthen

Improve the safety management level of emergency material reserves and enhance the protection against extreme climate events and climate disasters

By 2025, the average loss from meteorological disasters will account for 1.5% of the GDP of the region.

The proportion shall not exceed 0.15%.

Strengthen the ability to defend against extreme climate events. Gradually improve power supply, water supply, and drainage.

Establish and operate standards for urban lifeline systems such as water, gas, and communications. Strengthen public transportation

Climate risk assessment for safe operation of transportation facilities, including landslides, mudslides, floods, etc.

In areas prone to climate disasters, research and formulate emergency mechanisms to improve the ability to respond to extreme climate events capacity to ensure the smooth and safe operation of public transportation facilities. Encourage the construction and operation of power generation projects

Conduct climate change risk assessments during the process and improve the wind, pressure and frost resistance of power transmission and transformation facilities.

Standards and improve emergency response plans.

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Section 3 Deepening the reform and innovation of market-oriented mechanisms for responding to climate change in Guangdong

Fully cooperate with the construction of the national carbon market and combine the progress of the construction of the national carbon market to Further deepen the construction of Guangdong's carbon market, improve the total amount control mechanism of Guangdong's carbon market, promote the healthy and stable operation of Guangdong's carbon market and gradually make a smooth transition to the national carbon market. Establish a carbon credit system covering the whole province, actively develop green finance, and improve climate investment Financing mechanism.

1. Deepening the construction of Guangdong's carbon trading mechanism under the background of the national carbon market

Fully cooperate with the construction of the national carbon market. Further enhance the data of our province's power companies. Monitoring, data reporting and verification, quota allocation, transaction registration and settlement, market risk control The ability to implement and comply with the quota settlement agreement, strictly control the quality of carbon emission data, and establish data quality Regular inspection and random sampling are implemented to ensure that Cooperate with our province's power companies to do a good job in every compliance cycle of the national carbon market. In line with the progress of the national carbon market construction, we will prepare for the Guangdong carbon market to be included in the national carbon market enterprises The transition work of the industry should be strengthened, and the training of enterprises to be included should be strengthened. In terms of the capabilities and levels in all aspects of carbon trading, we should adapt to the requirements of the national carbon market as soon as possible. Promote the smooth connection of enterprises to be included in the national carbon market.

Further deepen the construction of Guangdong's carbon market mechanism. Formulate and issue Guangdong's carbon trading support Carbon peak and carbon neutrality implementation plan. Prepare the annual quota allocation plan for Guangdong carbon trading. The system comprehensively considers the requirements of high-quality economic development and reasonably sets the total annual quota and the The total amount of the elimination mechanism shall be optimized, and the factors such as the proportion of paid quota allocation, the amount of free quotas for emission-control entities, the baseline and the annual decline coefficient shall be optimized, and differentiated quota allocation cycles shall be explored.

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The industrial policy under the guidance of carbon peak and carbon neutrality will increase the proportion of paid quotas for new projects.

Further improve the construction of carbon emission monitoring, reporting and verification systems for emission-controlled enterprises, optimize the first

The three-party verification and review rules shall be established to establish a dynamic management mechanism for the management name of the third-party institution.

Guangdong's carbon market policy and system system promotes the "Guangdong Province Carbon Emission Management Trial Measures"

"Guangdong Province Carbon Emission Quota Management Implementation Rules"

Reporting Management Rules, Guangdong Provincial Enterprise Carbon Emission Verification Standards, Guangdong Provincial Carbon Emission Verification

Revision of the "Methods for Inspection and Review" etc.

Orderly expand the coverage of Guangdong's carbon market.

The peak strategy was organized to carry out enterprises in industries such as textiles, ceramics, data centers, transportation, and construction.

carbon emission accounting in the industry, gradually incorporate it into the carbon market, and promote the carbon market to become a

It is an important policy tool to achieve the carbon peak target in the construction and other fields.

Release characteristics, set differentiated entry thresholds for industries and enterprises, and gradually lower the entry thresholds.

Explore the inclusion of non-CO2 greenhouse gases such as methane in the carbon market coverage.

To build the foundation of the East China Carbon Market, study and build the Guangdong-Hong Kong-Macao Greater Bay Area Carbon Market, and promote investment from Hong Kong and Macao

Participants in the Guangdong carbon market will be allowed to trade and establish a cross-border trading mechanism for carbon emission rights.

2. Promoting the construction of a multi-level carbon market

Deepen Guangdong's carbon credit mechanism. Improve and issue the Guangdong Provincial Carbon Credit Trading Management Regulations.

The Carbon Credit Law and a series of supporting policy and technical document systems, including the Carbon Credit Methodology in various fields,

Optimize the application, issuance and transaction process of carbon credit projects. According to the principle of "joint construction, interconnection and sharing"

Based on the principle of

Encourage all sectors of society to develop and register carbon credit methodologies in the fields of new energy, energy conservation

and emission reduction, and natural resources. Support the development of basic

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Support Guangdong and other provinces in developing carbon credit mechanisms and product development.

Cooperation, establish a common carbon credit mechanism, and promote the mutual benefit of inter-provincial carbon credit emission reduction products

Improve the carbon inclusive business incentive mechanism and actively guide enterprises to low-carbon production and procurement.

Encourage residents to practice green travel, low-carbon living and consumption behaviors.

Strengthen linkage with voluntary emission reduction mechanism. Fully mobilize the whole society to participate in ecological environment protection.

The enthusiasm for conservation is encouraged, and enterprises, institutions and other entities are encouraged to develop forestry carbon sinks, marine carbon sinks,

Voluntary emission reduction projects such as photovoltaic power generation, low-carbon actions by residents, and resource recycling.

Carbon emission rights management and offset mechanism, dynamically adjust the carbon market to carbon quota within a reasonable range

Preferential emission reductions and the proportion of voluntary emission reduction projects, and encourage contract-compliant enterprises to purchase them first

Forestry carbon sinks and other carbon credits certified emission reductions in poverty-stricken areas.

Green Power Trading Rules (Trial)" to improve the green power certificate trading mechanism and do a good job

Green certificate trading is linked to the voluntary emission reduction mechanism.

3. Actively develop green finance

Deepen green finance reform and innovation. Coordinate the planning of green finance development in the province and actively

Promote the establishment and high-quality development of green finance reform and innovation pilot zones, and continuously enhance the "double

The radiation and driving capacity of green finance in the coastal economic zone, and the improvement of financial services for green industries in the coastal economic zone

The industrial development level and ecological products of the northern ecological development zone can better realize the economic value.

The construction of a green financial system encourages financial institutions to set up specialized institutions to serve green development.

Cultivate green intermediary service institutions, improve green financial infrastructure construction, and innovate green

Financial products and services will upgrade the industrial structure. Coordinate fiscal and financial policies in supporting green recovery and sustainable

development, and encourage local governments to participate in equity investment according to market principles.

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Provincial existing policy funds, or the establishment of green development guidance funds in a market-oriented manner in light of actual conditions, to support financial institutions in exploring forest carbon sink and marine carbon sink compensation incentive mechanisms.

Strengthen support for the development of green and low-carbon industries and technology research and development.

The investment and financing of high-energy-consuming projects such as electricity, steel, cement, and petrochemicals will be strengthened, and support will be increased for projects such as energy conservation and environmental protection, new energy, new energy vehicles, and carbon capture, utilization, and storage.

Establish a negative investment list to curb high-carbon investment and explore the use of investment subsidies, loan interest subsidies, etc.

Various means are used to innovate incentive and constraint mechanisms to promote corporate emission reduction.

promote mutual recognition of green finance standards among the three regions, and support enterprises to Hong Kong and Macao issue green bonds. Promote social responsibility information disclosure standards and guide listed companies Require bond issuers to disclose carbon emission information and encourage investors to make responsible investments.

Improve the climate investment and financing mechanism. Support Shenzhen in applying for the establishment of a national climate investment and financing promotion Support Shenzhen Futian District and Guangzhou Nansha New District to apply for the establishment of a local pilot project for climate investment and financing

Comprehensively use fiscal, tax, financial and differentiated regulatory means to formulate climate investment and financing

Encourage and guide social capital to enter the field of climate investment and financing,

Encourage enterprises and institutions to fully consider the impact of future market carbon prices in their investment activities.

Increase innovation in carbon finance products, optimize the carbon emission rights mortgage financing process, and expand carbon emission Decentralize the scale of mortgage financing. Improve the carbon quota spot forward business and encourage overseas entities

Participate in cross-border RMB payment channels. Comprehensively use a variety of policy incentive tools to enhance Increase participation in carbon financial products and continue to promote innovation in carbon financial products. Support Guangzhou Futures Exchange The exchange is accelerating the listing of strategic trading products such as carbon emissions and electricity that serve green development, exploring Carbon futures contract products with Guangdong Carbon Allowance (GDEA) as the underlying asset.

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(IV) Accelerate the establishment of Guangdong's carbon labeling mechanism

Accelerate the establishment of Guangdong Province's product carbon footprint evaluation and labeling (referred to as "Guangdong Carbon Label") Mechanism. Relying on the Guangdong Carbon Labeling Professional Committee to accelerate the construction of the Guangdong Carbon Labeling Mechanism, Formulate and publish the Guangdong Carbon Label Management Method and other document systems to standardize and supervise the Guangdong Province carbon label evaluation institutions, evaluation processes and label use. Establish a unified Guangdong carbon label technical specifications, evaluation procedures, identification and information management platform, research and construction Establish Guangdong carbon label evaluation technical standard system. Select Guangdong's superior products and encourage enterprises Track and manage product carbon footprints, and support those with the foundation, ability and willingness to Typical enterprises and products take the lead in participating in the Guangdong Carbon Label Evaluation and Application. Promote the companies and products with the label and tell the story of Guangdong companies that are responding to climate change story.

Promote international cooperation and mutual recognition of Guangdong carbon labels. Support third-party institutions to accelerate the expansion Green and low-carbon trade technical services, helping enterprises cope with green trade barriers and promoting the Australian technical institutions have established a technical standard system and implementation procedures for mutual recognition of carbon labels between Guangdong, Hong Kong and Macao. We will strive to complete the mutual recognition of carbon labels between Guangdong, Hong Kong and Macao by the end of 2025. On this basis, we will strengthen exchanges with the EU, the US, the UK and other countries in the field of carbon labeling and accelerate the promotion of Promote international cooperation and mutual recognition of carbon labeling.

Section 4: Promoting the modernization of climate change governance system and governance capacity

Adhere to the system concept, actively promote the construction of the policy system for addressing climate change, systematically strengthen the level of scientific and technological innovation in addressing climate change, and actively increase the external opening up in addressing climate change.

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We will improve the level of cooperation, comprehensively enhance the basic capabilities for addressing climate change, and effectively promote reform and exploration work to address climate change.

1. Establish and improve a management system for addressing climate change

Continue to improve the regulatory and standard system for addressing climate change. In the fields of ecological environment protection, resource and energy utilization, land space development, urban and rural planning and construction, etc., local regulations and standards In the process of drafting and revising the Regulations, we will promote the addition of content related to addressing climate change. The connection between the regulatory and standard systems, revision of the "Guangdong Province Carbon Emission Management Trial Measures", and completion of Improve the standards for carbon emission accounting in key industries, product carbon footprint assessment, and carbon emission information reporting Promote the formulation of carbon emission intensity standards for key industries and include construction projects that exceed the standards in the industry. Encourage industry associations and enterprises to conduct relevant research and publish carbon emission Release management group standards.

Improve environmental impact assessment, pollutant discharge permit system and clean production audit to promote the Incorporate greenhouse gas emission control and climate change response requirements into environmental impact assessments, and promote pollution The environmental impact assessment management of pollutants and greenhouse gas emissions is integrated and promoted in Guangdong Province. Pilot work on environmental impact assessment of greenhouse gas emissions from chemical industry construction projects, strictly control The greenhouse gas emissions of major projects in the petrochemical industry will be increased, and support will be provided to the power, steel, and construction industries. Key industries such as materials, nonferrous metals, petrochemicals, and chemicals are actively conducting research on the environmental impact of greenhouse gas emissions. Organize and carry out greenhouse gas emissions and pollution permitting in key industries. It can manage relevant research and promote key industries to carry out relevant research on pollutant and greenhouse gas emissions. Unified data collection, mutual supplementation and cross-checking. Carry out clean production audit Innovate pilot projects and strengthen the role of clean production in upgrading and transforming key industries and regions to reduce pollution and carbon emissions and in deepening the battle against pollution.

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Improve the greenhouse gas statistics, monitoring, accounting and reporting system. Improve the statistical system and assessment mechanism for carbon emission management in the province, and promote the coordinated development of environmental statistics.

Surveys on greenhouse gas emissions and exploring the use of management indicators related to addressing climate change as a

Regularly compile provincial and municipal greenhouse gas emission inventories, encourage prefecture-level cities to carry out pilot projects on county-level greenhouse gas inventories, and gradually expand the compilation of county-level inventories.

to improve the statistical and accounting work basis for greenhouse gases and explore ways to promote

Interoperability of data between departments. Standardize greenhouse gas emission reporting and verification activities, and gradually

Expand the scope of industry coverage and improve the accounting, reporting and verification system related to methane emission control.

Carry out pilot work on carbon monitoring and assessment, and explore the gradual inclusion of carbon monitoring into the ecological environment monitoring system

Coordinated implementation. Through satellite remote sensing, model analysis, flight monitoring, drone remote sensing, etc.

A variety of technical means are used to select sites for the urban greenhouse gas monitoring network, relying on existing communication towers,

Meteorological towers or a number of new high tower stations will be built to carry out automatic monitoring of major greenhouse gases.

Comprehensive analysis and application of greenhouse gas monitoring data, and quality control of major greenhouse gases

Research on technical methods of control business and strengthen the quality and traceability of greenhouse gas monitoring.

Strengthen the coordination of climate change response and ecological environmental protection systems.

Technical standards, data statistics and assessment mechanisms to build a coordinated control of greenhouse gases and

The management system of pollutant emissions, strengthening greenhouse gas and pollutant emissions, and adaptation

The synergy between climate change and ecological protection and restoration, the implementation of nature-based solutions, and the support of

Support the in-depth fight against pollution and the carbon dioxide peak action.

The same governance mechanism promotes unified policy planning standards, unified monitoring and evaluation, and unified supervision

Law enforcement, unified supervision and accountability.

Strengthen incentive and constraint economic policies. Implement environmental protection, energy conservation and water conservation, and new

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Improve the green charging price mechanism and establish a sound sewage and garbage treatment collection and pricing system.

The policy of charging tariffs, differentiated pricing policies and industry tiered pricing policies are promoting green, low-carbon and recycling

Establish and improve the ecological protection compensation mechanism and innovate the market-oriented mechanism of ecological protection

compensation. Explore the mechanism for realizing the value of ecological products and promote the protection of ecological resource rights.

Improve carbon emission rights trading mechanism, explore pilot projects for carbon sink rights trading.

The paid use system of emission rights will expand the types and transactions of pollutants in emission rights trading.

Improve green consumption price policies and green and low-carbon product support policies.

2. Strengthening scientific and technological innovation to address climate change

Accelerate the research and development of green and low-carbon cutting-edge technologies. Increase the research and development of key core low-carbon technologies.

Deploy a number of forward-looking, strategic low-emission technology R&D and innovation projects.

Energy conservation, resource recycling, new energy development, clean production technology and other fields

Key low-carbon technology innovation research, focusing on carbon dioxide capture, utilization and storage,

High-value conversion and utilization of carbon dioxide, key materials for reducing non-carbon dioxide greenhouse gas emissions, high-efficiency

Solar energy utilization, large-scale wind power, complementary utilization of wind and solar energy, advanced nuclear power, smart energy

Strengthen the cross-industry cooperation in new energy, new materials and new technologies.

Fusion research.

Strengthen the integration and application of green and low-carbon technologies and engineering. Study and formulate green technology targets

Catalogue of key energy-saving and low-carbon technology promotion, energy-saving and emission-reduction and low-carbon technology achievement transformation

Promote the list, build a green technology storage and transfer platform in the Guangdong-Hong Kong-Macao Greater Bay Area, and provide

Provide leading green technology transfer, transformation and industrialization comprehensive services in China. Establish and improve the

testing and evaluation system for green and low-carbon technologies and products, and promote technological innovation through market means.

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promote the application of pollution reduction and carbon reduction technologies, and vigorously cultivate green and low-carbon technology innovation.

New enterprise.

Strengthen basic scientific research on addressing climate change. Actively carry out research on the causes and

Adaptation, climate change analysis, assessment and risk assessment, carbon sinks in terrestrial and marine ecosystems, carbon dioxide capture, utilization and storage,

greenhouse gas monitoring and emission reduction, climate change

Basic scientific research in areas such as health risk assessment is encouraged to carry out the impact of climate change on Guangdong Province.

Water resources, agriculture, oceans and coastal areas, infrastructure and other key areas of impact, wind

Analysis and assessment of disaster risks and research on disaster risk management. Strengthen the foundation for synergistic efficiency of pollution reduction and carbon reduction.

Basic science and mechanism research, carry out analysis methods and follow-up evaluation of synergistic efficiency of pollution reduction and carbon reduction

and technology selection, and in national priorities such as air pollution prevention and control, carbon peak and carbon neutrality.

Set up research tasks in R&D projects, build a number of relevant key laboratories, and deploy implementation

A number of key innovation projects.

Strengthen scientific and technological innovation support in the field of climate change. Accelerate the training of talents and

Introduce and expand professional support teams for addressing climate change, implement high-level

Focus on the major scientific and technological needs of Guangdong, Hong Kong and Macao in achieving carbon peak and carbon neutrality

Seek, innovate scientific research mechanisms, build a low-carbon development scientific research platform, and promote the construction of Guangdong and Hong Kong

Promote the construction of collaborative innovation platforms for ecological and environmental protection science and technology such as the Australian Ecological and Environmental Science Center

As a national ecological environment innovation center, we will develop peak policy, climate governance policy, and industry

Development, technological innovation and application research.

3. Improving the level of openness and cooperation in addressing climate change

Strengthen climate governance cooperation in the Guangdong-Hong Kong-Macao Greater Bay Area. Improve the liaison and coordination mechanism for responding to climate

change in Guangdong, Hong Kong and Macao, and actively build a clean production technology research and development, promotion and financing mechanism in Guangdong, Hong Kong and Macao.

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system, explore the establishment of a green finance standard system for the Guangdong-Hong Kong-Macao Greater Bay Area, and promote the two regions in low-carbon development, carbon labeling, new energy vehicles, green buildings, green transportation, and meteorology. Information, urban adaptation to climate change and other fields.

Strengthen international exchanges and cooperation in addressing climate change. Promote cooperation with governments, enterprises and related organizations in advanced countries (regions) such as the European Union, the United Kingdom, and California, USA to address climate change. Exchange and cooperation on climate change, strengthen talent cultivation, green and low-carbon technology, and cooperation in equipment, projects, services and infrastructure, and strengthen financial connectivity, Actively participate in green and low-carbon international standard cooperation and strengthen International benchmarking for key industries and key products.

Promote green "Belt and Road" construction cooperation. Strengthen Guangdong Province's green development experience International promotion, expansion of cooperation mechanisms, promotion of low-carbon infrastructure, low-carbon industrial parks, Low-carbon energy, low-carbon transportation, low-carbon technology research and development, low-carbon product and service trade, gas pragmatic cooperation and coordinated development in the fields of climate finance, low-carbon talent training and exchanges. Encourage and guide enterprises to participate in the green "Belt and Road" construction, promote advanced low-carbon technologies, equipment, production capacity and adaptability of technology and products to "go out" and "bring in", improve the South To promote the effectiveness of South-South cooperation and promote the countries along the “Belt and Road” to actively respond to climate change.

Section 5: Implementing Multi-level Pilot Demonstration and Major Projects

Summarize and refine Guangdong's experience in low-carbon development that can be replicated and promoted, and focus on carbon peak We will carry out cutting-edge work such as carbon neutrality, near-zero carbon emissions, zero-carbon and negative carbon technology application demonstration, and adaptation to climate change, comprehensively deepen pilot demonstrations and major engineering construction to address climate change, and focus on

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Actively explore new paths for green and low-carbon development.

Continue to do a good job in low-carbon pilot construction. Select low-carbon development with good foundation and renewable energy. Regions with superior carbon source and carbon sink resources should take the lead in trial implementation, and actively explore pilot projects Explore new models and technologies for low-carbon development in the fields of industry, energy, transportation, construction, consumption, ecology, etc., and encourage selected pilot projects to implement clean energy substitution, resource recycling and utilization Use, carbon fixation, carbon capture and storage, purchase of voluntary emission reductions and other emission reduction and carbon sink enhancement measures. Explore diverse and differentiated pilot projects and accelerate the development of carbon-neutral products, services, activities, etc. Neutral accounting standards and offset mechanisms, accounting, recognition, purchase, Standardize the management of carbon offset and other processes, encourage and guide large-scale activities to follow the The "Guidelines for Carbon Neutrality Implementation (Trial)" took the lead in carrying out carbon neutrality practices. Zhongshan and other national low-carbon city pilot construction experience, Shenzhen International Low-Carbon City, Hengqin The Guangdong-Macao Deep Cooperation Zone New Area and other national low-carbon city (town) pilot projects are taking the lead. Internationally advanced level, further build a group of Establish pilot projects to explore new technologies, carbon sink capacity building and other fields. Establish an environment assessment and evaluation mechanism to timely grasp the progress of pilot implementation, summarize and promote advanced experience, Take rectification, adjustment, elimination and other measures against pilot entities with problems in the assessment.

Carry out collaborative innovation in pollution reduction and carbon reduction at the regional, urban, industrial park and enterprise levels New pilot projects are being launched. Encourage cities in the Pearl River Delta urban agglomeration and eastern, western and northern Guangdong to Cities and regions should speed up the exploration of effective models for synergistically increasing the efficiency of pollution reduction and carbon reduction, and explore different types of Urban pollution reduction and carbon reduction promotion mechanism, strengthen pollution reduction in urban construction, production and life Encourage all kinds of industrial parks to actively explore and promote the synergy of pollution reduction and carbon reduction based on their own leading industries and pollutant and carbon emission levels, and optimize the spatial layout of the parks.

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The Bureau vigorously promotes the use of new energy, promotes the optimization and cascade utilization of the park's energy system, the intensive and efficient recycling of water resources, the comprehensive utilization of waste, and the upgrading and transformation of sewage treatment. waste treatment facilities and waste incineration facilities to improve the level of green and low-carbon development of infrastructure. Key industry enterprises carry out pilot projects to reduce pollution and carbon emissions, and encourage enterprises to adopt measures such as process improvement, energy substitution, energy conservation and efficiency improvement, and comprehensive management to achieve air, gas and carbon reduction in the production process. The emissions of various pollutants such as water and solid waste as well as greenhouse gases have been significantly reduced, significantly improving the environmental protection Environmental governance performance, achieving pollutant and carbon emissions at the industry's advanced level; supporting enterprises The industry will further explore the path of deep pollution reduction and carbon reduction and create a benchmark enterprise with "double near-zero" emissions.

Carry out integrated demonstration projects of carbon capture, utilization and storage (CCUS) and the whole industry

Actively deploy basic scientific research and technological breakthroughs in carbon capture, utilization and storage We will tackle key problems and conduct research on using the submarine geological structure of the Pearl River Mouth Basin to store carbon dioxide. Identify and select sites that can be used for efficient, large-scale geological storage, and support large-scale, low-energy Actively promote the research and development of CCUS technology in the power industry Research, testing and commercial application, the steel industry carries out non-blast furnace ironmaking, hydrogen smelting, Pilot demonstration of low-carbon metallurgical technologies such as CCUS integration, and exploration of cement kiln tail The gas CCUS demonstration project, the petrochemical industry carried out green refining and CCUS integration demonstration, etc. Fan Engineering.

Promote the pilot work of climate change adaptation. Promote the demonstration of beautiful bay in Guangdong-Hong Kong-Macao Greater Bay Area. Model construction, explore the synergy between enhancing climate resilience and reducing pollution and carbon emissions. Promote Zhuhai's construction The pilot project of climate-adaptive cities is to carry out urban climate adaptation in typical climate-vulnerable areas such as the Pearl River Delta. The disaster prevention pilot project focuses on key areas such as ecological functional areas, coastal areas and islands, and has played a vital role in energy, transportation, construction and infrastructure safety, agricultural and forestry production, and marine economic development.

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Pilot work is being carried out in key areas such as development.

Actively implement key project construction. Organize in energy, industrial structure, resource conservation

Intensive utilization, ecological protection and restoration, scientific and technological innovation, green finance and other fields

Construction of large-scale projects, active mitigation and adaptation to climate change, promotion of green and low-carbon transformation of social and

economic development, and assistance in the implementation of carbon peak action.

Column Demonstration Project Construction
1. Low-carbon pilot demonstration projects. Carry out pilot demonstration projects for carbon peak in key industries such as industry, agriculture and rural areas, urban and rural construction, and transportation, and carry out the construction of the first batch of carbon neutral pilot demonstration cities (districts) in Shaoguan City, Shenzhen Qianhai Cooperation Zone, Shenzhen Longgang District, Guangzhou Conghua District, Guangzhou Huadu District, Zhuhai Hengqin Guangdong-Macao Deep Cooperation Zone, Zhongshan Cuiheng New District, Zhongshan Shenwan Town, and Shantou Nan'ao County. Implement demonstration projects for greenhouse gas control and emission in industrial production processes and demonstration projects for the development and utilization of agricultural climate resources. Build near-zero emission demonstration areas, including pilot demonstrations of low-carbon cities, low-carbon towns, low-carbon parks, and low-carbon communities, and implement near-zero carbon emission demonstrations in towns, buildings, transportation, communities, parks, and enterprises. 2. Pilot projects for coordinated innovation in pollution reduction and carbon reduction. Based on the goal of fighting a tough battle against pollution and reaching carbon peak, pilot projects for coordinated innovation in pollution reduction and carbon reduction are carried out at multiple levels in key areas, cities, industrial parks, and enterprises to help achieve the goal of green and low-carbon development. 3. Demonstration projects for comprehensive utilization of resources. Promote the construction of the Guangdong-Hong Kong-Macao Greater Bay Area into a "waste-free pilot zone." promote Cell projects such as "waste-free parks" and "waste-free communities" Support the construction of the "waste-free new district" in Cuiheng New District, Zhongshan. 4. Demonstration project for adapting to climate change. Carry out urban climate disaster prevention and control in typical climate-vulnerable areas such as the Pearl River Delta. Pilot projects: Implement the Guangdong Province drought resistance capacity improvement construction project and build the Guangdong Province emergency backup water source project. 5. Carbon capture, utilization and storage demonstration projects. Accelerate the research on carbon storage technology, explore the establishment of a technical system for storing carbon dioxide using submarine geological structures, and promote related industrialization. Carry out in-depth testing of carbon dioxide capture technology, and carry out full-process demonstration projects and industrialization construction of carbon dioxide capture, utilization and storage in the power, cement, chemical and other industries. Accelerate the feasibility study of the 10 million ton demonstration project of the Daya Bay CCUS industrial cluster. 6. Carbon monitoring network construction pilot project. Set up atmospheric greenhouse gas monitoring stations in Guangzhou, Shenzhen and Shaoguan, and configure monitoring instruments such as CO_2, CH_4, and CO to understand the greenhouse gas concentration in pilot cities and estimate urban carbon emission fluxes. Carry out carbon sink monitoring of mangrove ecosystems in Shenzhen and Zhanjiang, and carry out carbon sink monitoring of typical forest ecosystems in Shaoguan to estimate ecosystem carbon reserves.

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Section 6. Safeguard Measures

Strengthen organizational implementation. Implement the responsibility list for addressing climate change and establish and improve provincial The departments shall promote the division of labor and cooperation mechanism of this plan. The local governments shall, based on the target indicators and main tasks determined in this plan, combine the actual situation and scientifically and reasonably break down the target tasks. To coordinate and supervise the work arrangements, clarify the responsible units for each task, Responsible personnel and target completion deadlines, ensuring that responsibilities, measures and investments are in place, Ensure that all work is carried out effectively and orderly.

Strengthen overall coordination. Fully mobilize the resources of each department to carry out Strengthen the city-level planning and relevant departments The special plans in relevant fields should be linked with the provincial special plans for addressing climate change to ensure the Strengthen the coordination and cooperation in the field of climate change Coordinate and cooperate on fiscal, tax, financial, price, land, industry and other policies to support climate change Implementation of relevant policies will create a comprehensive synergy.

Provide financial support. Broaden investment and financing channels and give full play to the guiding role of fiscal funds. Establish and improve a normalized and stable fiscal system. Funding mechanism, under the premise of firmly curbing the increase of hidden debt, explore various By combining local finance with social funds, we will actively guide private capital to participate in the and climate change response projects, ensuring various pilot demonstrations, green technology research and development and Investment in promotion, capacity building, etc.

Strict supervision and assessment. Improve the planning implementation evaluation indicator system and establish a scientific and reasonable A dynamic planning implementation evaluation mechanism will be established to strengthen the monitoring and scheduling of the implementation of this plan.

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Establish and improve scientific and comprehensive assessment systems and methods, strengthen the target responsibility system for controlling greenhouse gas emissions, and strengthen the application of assessment results.

The completion of greenhouse gas emission control targets will be included in the relevant assessment of the ecological environment, and gradually formed An ecological environment assessment system that reflects the requirements for synergistic efficiency improvement of pollution reduction and carbon reduction.

Provide good publicity and guidance. Summarize advanced examples of addressing climate change and “zero-carbon” development. models and successful cases, making full use of the national "Belt and Road" government cooperation, climate change Through platforms such as South-South cooperation, we actively spread and share Guangdong’s experience in low-carbon transformation of the economy and society.

Organize and carry out the "June 5th" Environment Day, Ecological Civilization Construction Publicity Week, Energy Conservation Publicity Week, National Low Carbon Day, World Meteorological Day, National Disaster Prevention and Reduction Day and other themed publicity activities were held to promote Actively create a social atmosphere of energy conservation, low carbon and green life.